

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)							Semester: II		
Course: Basic Electrical Engineering							Code: ECES203		
Teaching Scheme (Hrs/week)				Evaluation Scheme (Marks)					
Lecture	Practical	Tutorial	Credit	CIE	ETE	TW	OR	PR	Total
03	02	-	03	40	60	25	-	-	125
Prerequisites:									
Fundamental Electrical Concepts, Familiarity with Electrical Units & basic safety precautions.									
Course Objectives:									
1. To familiarize students with the fundamentals of Electrical Engineering.									
2. To make students aware about the functioning of electrical machines, batteries and their applications.									
3. To introduce students to the components of low-voltage electrical installations and the methodology for estimating energy bills.									
Course Outcomes: After completion of this course, students will able to -									
CO1	Understand work, power, and energy relationships, unit conversions, and Lead Acid and Lithium-Ion battery charging/discharging processes.								
CO2	Analyze simple resistive circuit powered by DC supply using circuit theorems.								
CO3	Interpret voltage, current, phase relationship for RLC loads.								
CO4	Examine voltage, current and power relationships in star and delta AC circuits, including protection systems.								
CO5	Explain operational principle of transformer, DC machines and induction motor.								
CO6	Estimate the energy bill for domestic consumers.								
Course Contents:									
Unit	Description								Duration (Hrs.)
1.	Work, Power and Energy and Batteries: Basic Definition and Units: Length, Mass, Time, Temperature, Area, Volume, Acceleration, Density, Velocity, Pressure, Work, Energy, Torque, Power, Voltage, Current, Resistance, Capacitance, Conductance, Charge, Inductance, Frequency, Impedence etc., Multiples and Submultiples, Types of units (MKS, CGS and SI), Unit conversions. Work, Power and Energy: Effect of temperature on resistance, resistance temperature coefficient (derivation and numerical), insulation resistance of single core cable (derivation and numerical), conversion of energy from one form to another in electrical, mechanical, and thermal systems. Batteries: Lead acid and Lithium Ion battery – (Construction, working, charging and discharging and its applications),concept of depth of charging, state of charge of battery, battery capacity, battery efficiency, ampere-hour and watt-hour of battery, maintenance of batteries, and series-parallel connection of batteries.								7
2.	DC Circuits: Analysis of series and parallel circuits, KVL and KCL (statement, sign convention), ideal and practical voltage and current sources, simple mesh and node analysis, source transformation (simple numerical), star-delta transformation (simple								7

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	numerical), Superposition and Thevenin's theorem (Statement and numerical - only for independent sources, and resistive circuit).	
3.	AC Circuits (Single phase circuits): Generation of sinusoidal voltage, representation of sinusoidal waveforms, concept of cycle, period, frequency, instantaneous, peak, average, and RMS values, Lagging, leading and in phase quantities and their phasor representation, Rectangular and polar representation of phasors, Concept of real, reactive, apparent, complex power and power factor, Analysis of single-phase AC series circuit (pure R, L, C and series R-L, R-C, and R-L-C combinations), Concept of impedance, admittance, voltage-current, power waveforms and relevant phasor diagrams for different combinations.	7
4.	AC Circuits (Three-phase circuit) and Electrical Installations: AC Circuits (Three-phase circuit): Concept of three-phase supply and phase sequence, Three-phase balanced circuits, voltage and current relations in star and delta connections, and power calculations. Electrical Installations: Components of LT Switchgear: Fuse, MCB, MCCB (working, advantages, disadvantages and applications), Earthing - (Definition, importance of earthing, types, advantages of earthing, difference between earthing and neutral).	7
5.	Single Phase Transformer: Construction, working principle and EMF equation of transformer, Ideal and practical transformer, Losses, Types of transformers (Step up and step-down transformer), Concept of voltage regulation and efficiency (simple numerical), Introduction to auto-transformer (Construction, working, advantages and applications). Electricity Bill: Power rating of household appliances, Definition of "unit" used for consumption of electrical energy, Two-part electricity tariff, Calculation of electricity bill for domestic consumers.	7
6.	DC Machines: DC generator and motor (Construction, working principle, types, and applications), emf equation of DC generator, (Simple numerical). Voltage expression of generator and motor (Simple numerical), Concept of back-emf (simple numerical), Armature and shaft torque equation (only descriptive treatment). AC Machines: Constructional features, working principle of three-phase induction motor, types (squirrel cage and slip ring), concept of synchronous speed, rotor speed, slip, power stages in three phase induction motor, concept of torque equation of three phase induction motor, torque-slip characteristics, industrial applications of induction motor.	7
TOTAL		42
List of Experiments:		
Group A: Minimum SIX experiments from following list - Demonstration of measurement of various units and their conversions. - Measurement of insulation resistance of electrical equipment/cable using Megger. - Verification of Superposition theorem with DC supply using hardware. - Verification of Thevenin's theorem with DC supply using hardware. - Measurement of steady state response of series RL and RC circuits on AC supply and observations of voltage and current waveforms on storage oscilloscope. - Verification of relation between phase and line quantities in three-phase balanced star and delta connections of load.		

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7. Demonstration of different types of electrical protection equipment such as fuses, MCB, MCCB and earthing system.
8. Calculation of efficiency and voltage regulation of single-phase transformer by direct loading test.
9. Demonstration of cut-out sections of machines: DC machine (commutator-brush arrangement), Induction machine (squirrel cage rotor).
10. Analysis of LT electricity bills and energy conservation (Case study).

Group B: DIY Models – Any TWO, from following list or any other suitable model

11. Demonstration of fundamental laws of Electrical Engineering using breadboard – (a) Ohm's Law (b) Faraday's law of Electromagnetic Induction (c) Kirchhoff's laws (KCL and KVL).
12. Generation of power with magnets and copper wire.
13. Conversion of mechanical energy to electrical energy and vice versa.
14. Model of mutual induction in transformer.

Group C: A mandatory visit to any transformer/electrical machines manufacturing industry.

Text Books:

1. B.L. Theraja, A. K. Theraja, "A Textbook of Electrical Technology" - Volume I: Basic Electrical Engineering", S. Chand Publication.
2. V. K. Mehta, Rohit Mehta, "Basic Electrical Engineering", S. Chand and Company Private Ltd.
3. D. P. Kothari, I.J. Nagrath, "Theory and Problems of Basic Electrical Engineering", PHI Publication.
4. Bharti Dwivedi, Anurag Tripathi, "Fundamental of Electrical Engineering", Wiley Publication.

Reference Books:

1. E. Hughes, "Electrical and Electronics Technology", Pearson Publication.
2. L. S. Bobrow, "Fundamentals of Electrical Engineering", Oxford University Press.
3. H Cotton, "Electrical technology", CBS Publications.
4. D. C. Kulshreshtha, "Basic Electrical Engineering", McGraw Hill.

E-Resources:

1. A NPTEL Course on "Fundamentals of Electrical Engineering", IIT Khargpur - <https://archive.nptel.ac.in/courses/108/105/108105112/>
2. A NPTEL Course on "Basic Electrical Technology", IISc Bangalore - <https://archive.nptel.ac.in/courses/108/108/108108076/>
3. Virtual lab – Amrita Vishwa Vidyapeetham - <https://vlab.amrita.edu/?sub=1&brch=75>
4. Electrical Engineering Basics - <https://www.classcentral.com/classroom/youtube-electrical-engineering-basics-54532>